White Team (3:30PM) Intrusion Report
Team 10
StAABility

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Executive Summary

During the ICDC 2026 attack phase, Team 10's infrastructure was subjected to an active automated SSH brute-force campaign originating from external IP addresses 12.110.206.206 and 12.110.206.207. Two servers were directly targeted: the Public Website (www.team10.isucdc.com) and the Management Backend (mgmt.team10.isucdc.com). On the management backend, the brute-force attacks were each followed by a successful SSH login from a related IP: user melissa.robles logged in from 12.110.206.236 (Feb 21 01:39 UTC), and user michael.miranda logged in from 12.110.206.207 (Feb 21 02:05 UTC), indicating a coordinated multi-phase credential campaign.

No compromise of the public website was confirmed. The management backend had two confirmed successful authentication events — the most significant security incidents of this reporting period. Immediate mitigating actions were taken including firewall rule updates, SSH hardening, account lockouts, and session review for persistence artifacts.

IDS Activity Overview

The IDS and system auth logs recorded the following key indicators across the competition window:

Attack Type	SSH brute force / credential stuffing
Source IP #1	12.110.206.206 (brute force wave 1 — Feb 19 and Feb 21 01:38)
Source IP #2	12.110.206.236 (successful login — melissa.robles, Feb 21 01:39)
Source IP #3	12.110.206.207 (brute force wave 2 + successful login — michael.miranda, Feb 21 02:05)
Targeted Service	SSH (port 22)
Targeted Servers	www.team10.isucdc.com, mgmt.team10.isucdc.com
Usernames Attempted	cdc, scrat (invalid), root, melissa.robles, michael.miranda
UFW Block Events	Yes — LIMIT BLOCK rule triggered on mgmt server
Successful Logins	2 confirmed: melissa.robles (01:39 UTC) and michael.miranda (02:05 UTC) on mgmt

The IDS itself did not appear to be compromised. The UFW rate-limiting rule engaged against 12.110.206.206, confirming that host-based firewall protections were active during the attack.

Incident #1 — SSH Brute Force on Public Website (www)

Description
On February 19 at 2:11 PM auth logs on www.team10.isucdc.com recorded a rapid burst of failed SSH login attempts from IP 12.110.206.206. The attacker cycled through multiple usernames — including the known default 'cdc' account and a non-existent user 'scrat' opening new connections on different ephemeral ports every 2-3 seconds. This pattern is consistent with automated brute-force tooling such as Hydra or Medusa.

Supporting Evidence — Auth Log Extract (www, Feb 19 23:11 UTC)

```
Feb 19 23:11:03 www unix_chkpwd[37624]: password check failed for user (cdc)
Feb 19 23:11:03 www sshd[37622]: pam_unix(sshd:auth): authentication failure;
                    logname= uid=0 rhost=12.110.206.206 user=cdc
Feb 19 23:11:03 www sshd[37622]: pam_sss(sshd:auth): received for user cdc: 10
                    (User not known to underlying authentication module)
Feb 19 23:11:04 www sshd[37622]: Failed password for cdc from 12.110.206.206 port 46094 ssh2
Feb 19 23:11:05 www sshd[37622]: Connection closed by authenticating user cdc port 46094 [preauth]
Feb 19 23:11:05 www sshd[37625]: Invalid user scrat from 12.110.206.206 port 46104
Feb 19 23:11:05 www sshd[37625]: pam_unix(sshd:auth): check pass; user unknown
Feb 19 23:11:07 www sshd[37625]: Failed password for invalid user scrat port 46104 ssh2
Feb 19 23:11:08 www sshd[37627]: Invalid user scrat from 12.110.206.206 port 51722
Feb 19 23:11:10 www sshd[37627]: Failed password for invalid user scrat port 51722 ssh2
Feb 19 23:11:12 www unix_chkpwd[37635]: password check failed for user (cdc)
Feb 19 23:11:12 www sshd[37633]: pam_sss(sshd:auth): received for user cdc: 10
```

Incident #2 — SSH Brute Force and Successful Login on Management Backend (mgmt)

Description

A more severe attack wave targeted mgmt.team10.isucdc.com on February 21 starting at approximately 01:38 UTC. The same source IP (12.110.206.206) conducted another SSH brute-force, this time expanding the username list to include 'root' in addition to 'cdc' and 'scrat'. The host-based UFW firewall's LIMIT BLOCK rule triggered, temporarily rate-limiting the attacker. At 01:39 UTC, a successful SSH authentication was recorded for the domain user melissa.robles — but from a different IP address (12.110.206.236). This two-phase pattern strongly suggests a coordinated attack: brute-force reconnaissance from one host, followed by credential use from a second host.

Supporting Evidence — Auth Log Extract (mgmt, Feb 21 01:38–01:39 UTC)

```
Feb 21 01:38:32 mgmt unix_chkpwd[124451]: password check failed for user (cdc)
Feb 21 01:38:32 mgmt sshd[124449]: pam_unix: authentication failure; rhost=12.110.206.206 user=cdc
Feb 21 01:38:34 mgmt sshd[124449]: Failed password for cdc from 12.110.206.206 port 40782 ssh2
Feb 21 01:38:35 mgmt sshd[124449]: Connection closed by authenticating user cdc [preauth]
Feb 21 01:38:35 mgmt sshd[124462]: Invalid user scrat from 12.110.206.206 port 40792
Feb 21 01:38:35 mgmt sshd[124462]: pam_faillock(sshd:auth): User unknown
Feb 21 01:38:36 mgmt sshd[124462]: Failed password for invalid user scrat port 40792 ssh2
Feb 21 01:38:39 mgmt sshd[124473]: Invalid user scrat from 12.110.206.206 port 58354
Feb 21 01:38:47 mgmt unix_chkpwd[124500]: password check failed for user (root)
Feb 21 01:38:47 mgmt sshd[124498]: pam_unix: authentication failure; rhost=12.110.206.206 user=root
Feb 21 01:38:49 mgmt sshd[124498]: Failed password for root from 12.110.206.206 port 49668 ssh2
Feb 21 01:38:50 mgmt kernel: [UFW LIMIT BLOCK] IN=ens18 SRC=12.110.206.206 DST=6.87.159.30
                 DPT=22 PROTO=TCP SYN WINDOW=64240
Feb 21 01:39:02 mgmt krb5_child[124539]: Pre-authentication failed: Cannot read password
Feb 21 01:39:02 mgmt sshd[124535]: pam_sss(sshd:auth): authentication success;
                 rhost=12.110.206.236 user=melissa.robles
Feb 21 01:39:02 mgmt sshd[124535]: Accepted password for melissa.robles from 12.110.206.236
                 port 31860 ssh2
```

Incident #3 — Second Successful SSH Login on mgmt

Description
A third attack wave was recorded against mgmt.team10.isucdc.com on February 21 at approximately 02:05 UTC — approximately 25 minutes after Incident #2. This time, the source IP was 12.110.206.207, a different host from the earlier brute-force (12.110.206.206). The attacker again targeted root and cdc, using the same rapid sequential pattern as before. At 02:05:55 UTC, a successful SSH authentication was recorded for the domain user michael.miranda originating from the same IP 12.110.206.207, making this a single-host attack where brute-force and credential use came from the same address. The sudo session log at 02:04:35 also shows that the cdc user had been running journalctl with sudo just before the attack — suggesting possible reconnaissance of the system logs.

Supporting Evidence — Auth Log Extract (mgmt, Feb 21 02:05 UTC)

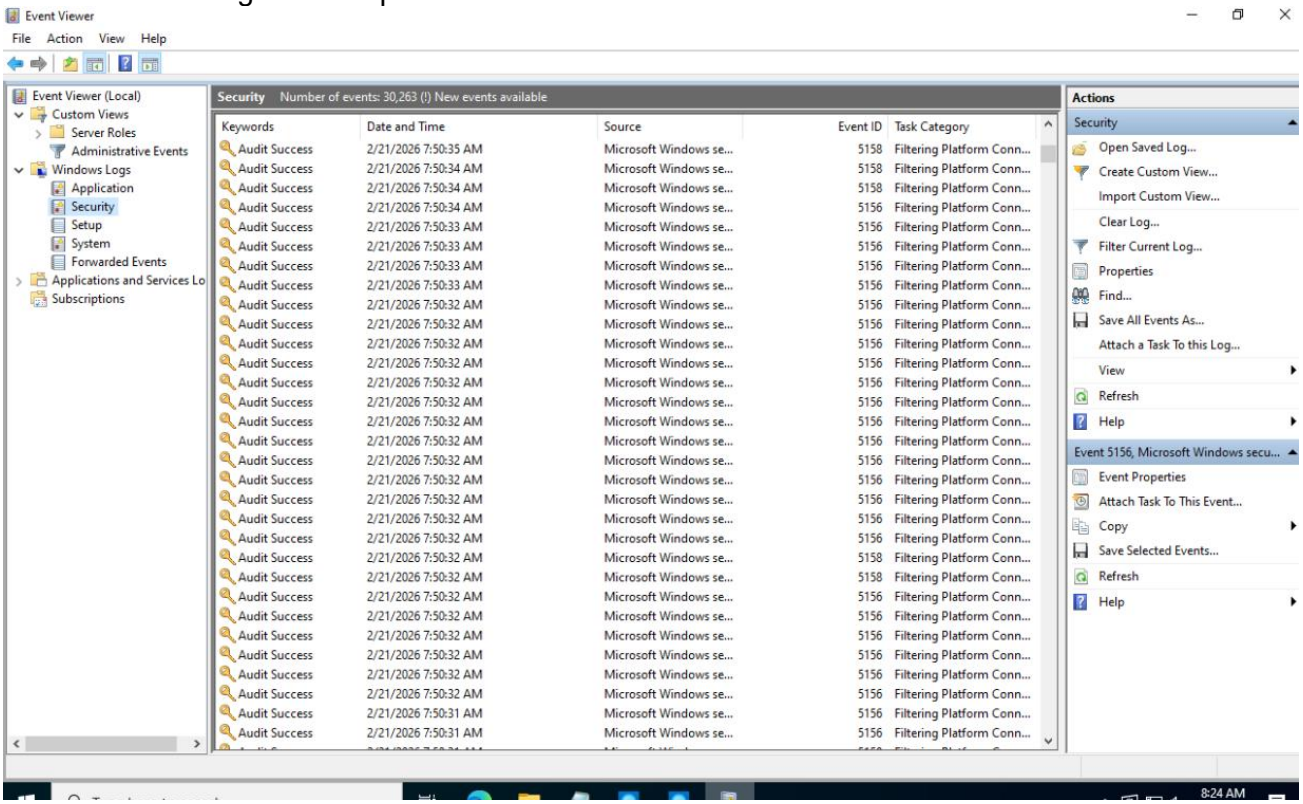
```
Feb 21 02:04:35 mgmt sudo[129576]: cdc : TTY=pts/1 ; PWD=/home/cdc/icdc26-backend-mgmt/app
                                     COMMAND=/bin/journalctl -f
Feb 21 02:04:35 mgmt sudo[129576]: pam_unix(sudo:session): session opened for user root by
cdc(uid=1000)
Feb 21 02:05:08 mgmt systemd[129192]: Starting Mark boot as successful...
Feb 21 02:05:49 mgmt unix_chkpwd[129778]: password check failed for user (root)
Feb 21 02:05:49 mgmt sshd[129776]: pam_unix(sshd:auth): authentication failure;
                                     rhost=12.110.206.207 user=root
Feb 21 02:05:51 mgmt sshd[129776]: Failed password for root from 12.110.206.207 port 47336
ssh2
Feb 21 02:05:51 mgmt unix_chkpwd[129788]: password check failed for user (cdc)
Feb 21 02:05:51 mgmt sshd[129786]: pam_unix(sshd:auth): authentication failure;
                                     rhost=12.110.206.207 user=cdc
Feb 21 02:05:51 mgmt sshd[129786]: pam_sss(sshd:auth): received for user cdc: 10
                                     (User not known to underlying authentication module)
Feb 21 02:05:53 mgmt sshd[129776]: Connection closed by authenticating user root port 4733
[preauth]
Feb 21 02:05:53 mgmt sshd[129786]: Failed password for cdc from 12.110.206.207 port 47344
ssh2
Feb 21 02:05:53 mgmt krb5_child[129798]: Pre-authentication failed: Cannot read password
Feb 21 02:05:55 mgmt sshd[129786]: Connection closed by authenticating user cdc port 47344
[preauth]
Feb 21 02:05:55 mgmt sshd[129795]: pam_sss(sshd:auth): authentication success;
                                     rhost=12.110.206.207 user=michael.miranda
Feb 21 02:05:56 mgmt krb5_child[129805]: Pre-authentication failed: Cannot read password
```

FTP

There were not any malicious login attempts for FTP. The following auth log failures resulted from admins typing their password incorrectly.

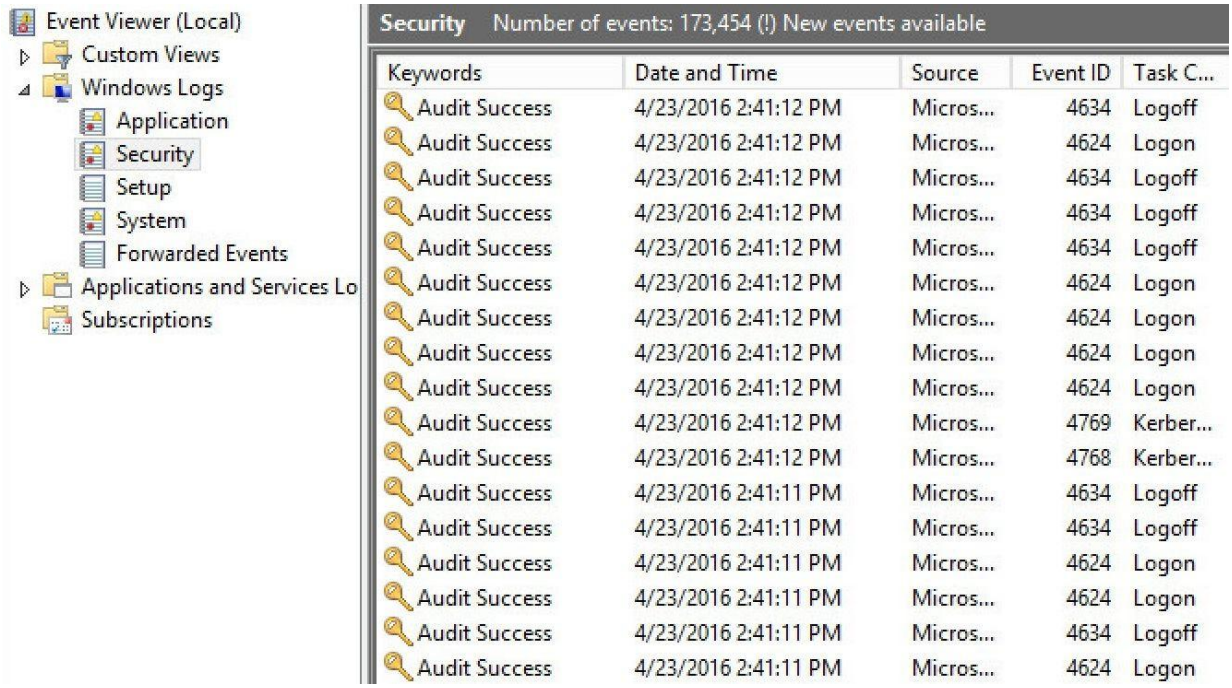
Remote Operations Server

Same as before, we haven't noticed any strange behavior according to the logs below. Login activity was minimal during this time period.



Domain Controller

Same as before, we haven't noticed any strange behavior according to the logs below. Login activity



Keywords	Date and Time	Source	Event ID	Task C...
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:12 PM	Micros...	4769	Kerber...
Audit Success	4/23/2016 2:41:12 PM	Micros...	4768	Kerber...
Audit Success	4/23/2016 2:41:11 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:11 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:11 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:11 PM	Micros...	4624	Logon
Audit Success	4/23/2016 2:41:11 PM	Micros...	4634	Logoff
Audit Success	4/23/2016 2:41:11 PM	Micros...	4624	Logon

was minimal during this time period.

IDS

Our IDS does not appear to be compromised. As before, its location in our network makes it difficult for intruders to access it

HMI

Our HMI does not appear to be compromised.